

Customer Shopping Behavior Analysis

Business Problem Statement & Project Deliverables

Dataset Overview

Based on the dataset containing **3,900 customer records** and **18 columns**, this project focuses on **Customer Shopping Behavior Analysis**. The dataset includes customer demographics, purchase history, product information, discounts, subscriptions, shipping methods, payment methods, and review ratings, making it well suited for retail business analytics.

Business Problem Statement

A retail company wants to better understand customer shopping behavior to improve sales performance, increase customer satisfaction, optimize marketing campaigns, and retain loyal customers. Although the company collects large volumes of customer transaction data, it lacks actionable insights into purchasing patterns, customer preferences, subscription behavior, product performance, and revenue generation. The objective of this project is to analyze customer shopping behavior using Python, MySQL, and Power BI. The project transforms raw transactional data into meaningful business insights through data preprocessing, exploratory data analysis (EDA), SQL-based business analysis, and interactive dashboards that support data-driven decision-making.

Business Objectives

- Understand customer purchasing behavior.
- Identify high-value customer segments.
- Analyze the impact of discounts and promotions.
- Evaluate subscription performance.
- Identify top-performing product categories.
- Improve customer retention strategies.
- Support business decisions with interactive dashboards.

Project Deliverables

- Data Cleaning & Preprocessing using Python.
- Exploratory Data Analysis (EDA).
- Feature Engineering.
- MySQL database creation and SQL business queries.
- Interactive Power BI dashboard.
- Business insights and recommendations.
- Final project documentation and report.